ZCAS University

CCS 1501 ENGINEERING MATHEMATICS

MID SEMESTER EXAMINATION

20TH OCTOBER, 2023

08:30 HRS-11:30HRS

TIME ALLOWED:

- **WRITING – THREE HOURS**
- **READING – 5 MINUTES**

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A (Compulsory)

QUESTION 1

(a) Consider a set $A = \{a, b, c\}$.

(i) Find the total number of subsets for a set A. [2]

(ii) List $P(A)$. [3]

(b) Simplify each of the following:

(i) $[A^c \cup (A \cap B^c)]^c$ [4]

(ii) $(A \cap B) \cup (A - B)$ [4]

(c) Show that each of the following numbers are rational.

(i) $4.\overline{82}$ [3]

(ii) $0.2\overline{13}$ [4]

(d) Let $\mathbb{R}$ be the universal set, $A = \{ x \in \mathbb{R}: -5 \leq x < 4\}$ and $B = \{ x \in \mathbb{R}: -1 \leq x\}$. Find and display the solution on the real number line:

(i) $A \cap B$ [5]

(ii) A^c [4]

(e) (i) Evaluate $f(0)$ and $f(-2)$ for each of the following functions.

(a) $f(x) = -2x + 5$ [1]

(b) $f(x) = |-2x + 5|$ [2]

(ii) Sketch the graph of each of the functions in (i) on the same axes. [7]

[Total: 40 marks]

SECTION B (Answer any THREE (3))

QUESTION 2

(a) Let $A = \left\{-4, \frac{1}{4}, -\sqrt{2}, 0.3, \frac{1}{\sqrt{3}}, 0.\overline{3}, \sqrt{2}, 3\right\}$. List the elements of the following subsets of A.

(i) Natural numbers (ii) Integers (iii)Rational numbers (iv) Irrational numbers (v) Real numbers. [5]

(b) Solve for x and y given that:

(i) $(x + iy)(4i) = 8$ [4]

(ii) $\frac{1-i}{x+iy} + \frac{1}{1+3i} = 1$ [5]

(c) If $X \subset Y$, then simplify the following:

(i) $X \cap Y$ [3]

(ii) (ii) $X \cup Y^c$ [3]

[Total: 20 marks]

QUESTION 3

(a) Given that a function $g(x) = x^2 - 6$, find:

i. $g(4)$ [1]

ii. $g(-4)$ [1]

iii. $g(2)$ [1]

iv. the possible real values of a is $g(a) = a$ [3]

(b) Express each of the following as the simplest possible surd:

(i) $\sqrt{48}$ [1]

(ii) $\sqrt{132}$ [1]

(iii) $\sqrt{88}$ [2]

(c) Rationalise the denominator:

(i) $\frac{2}{\sqrt{3}}$ [3]

(ii) $\frac{2}{2-3\sqrt{2}}$ [3]

(iii) $\frac{2}{(2-2\sqrt{3})(5-\sqrt{2})}$ [4]

[Total: 20 marks]

QUESTION 4

(a) Consider the function $f(x) = -2x^2 - x + 3$

(i) Explain whether the function has a maximum or minimum [2]

(ii) Determine the coordinates of the turning point. [2]

(iii) Determine the coordinates of the x and y intercepts [3]

(iv) Hence or otherwise sketch the graph of $f(x)$ [2]

(b) Given that $\mathbb{R}$, the set of real numbers is the universal set, $A = (-8,6]$ and $B = [5,\infty)$, find

(i) $A - B$ [2]

(ii) $B \cap A^c$ [3]

(c) Give a mathematical definition of $|x|$, the modulus of x. Hence or otherwise find:

(i) $\left|-\frac{2}{3} - 1\right|$

(ii) $|-2-x|$, when $x = -3$ [6]

[Total: 20 marks]

QUESTION 5

(a) Given that $z = 2 - i$ and $w = 3 + 4i$. Express each of the following in the form $a + ib$ where a and b are real numbers:

(i) $z - w$ [2]

(ii) zw [3]

(iii) $\frac{w}{z}$ [4]

(b) Discuss the function $f(x) = |x^2 - 2x + 1|$ [7]

(c) Write each of the following in its simplest terms:

(i) i^{45} [2]

(ii) $\frac{1}{i^{18}} - 3i + \frac{i^{24}}{2}$ [2]

[Total: 20 marks]

END OF EXAMINATION